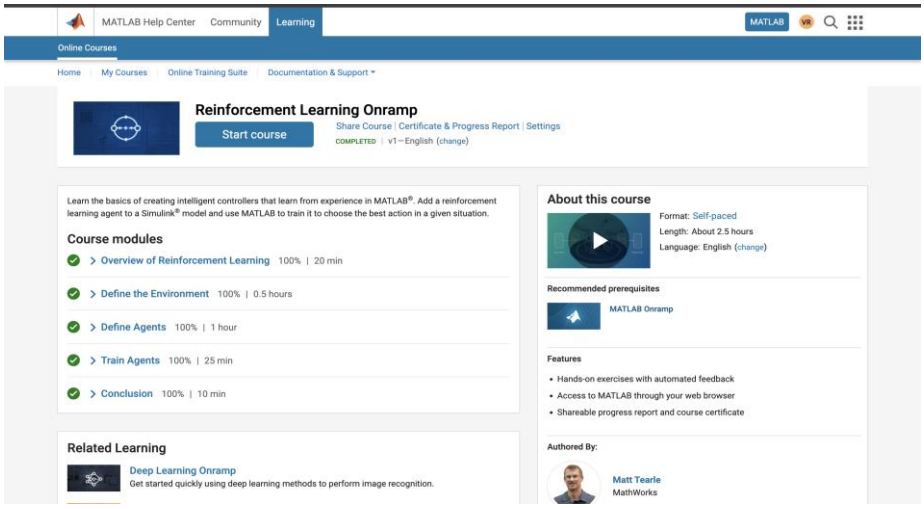
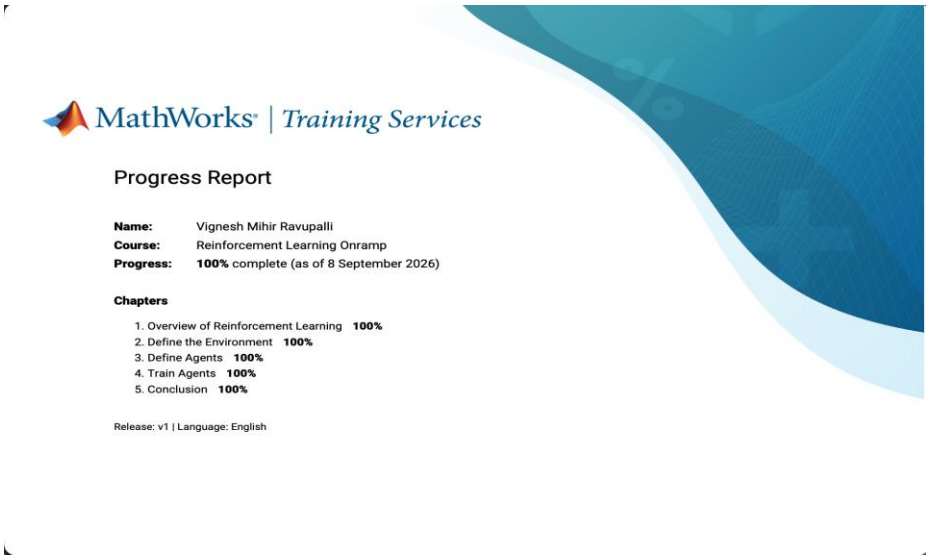
Task 1: Complete MATLAB Reinforcement Learning Onramp







Task 2: RL Agent Training Observation Table

Parameter	Observation
RL Environment	Simulink-based targetDice environment
Agent Used	rlQAgent (Q-Learning Agent)
Number of Training Episodes	Maximum: 2000; Actual completed: **886 episodes**
Initial Performance	Episode rewards were highly variable and generally low, with several rewards below -10 and occasional values reaching approximately -20
Final Performance	Final Episode Reward: **-4**; Final Average Reward: **-2.99**
Training Time	Approximately **3 minutes 20 seconds**
Training Stopped/Converged At	**Episode 886**, when the Average Reward reached the stopping criterion of approximately -3

Additional Training Settings Used

Training Parameter	Value
Max Episodes	2000
Max Steps Per Episode	20
Stop Training Criteria	AverageReward
Stop Training Value	-3
Score Averaging Window Length	100

Task 2: Answers to Questions

20. What type of RL environment is used in the Onramp exercise?

The exercise uses a **Simulink-based reinforcement learning environment** called targetDice, which interacts with the RL agent through the Simulink model.

21. What type of agent is trained?

An **rlQAgent**, which is a Q-learning reinforcement learning agent, is trained.

22. What is the purpose of training the agent?

The purpose is to enable the agent to learn an effective policy for selecting actions in the environment so that it can maximize its cumulative reward and perform the dice task successfully.

23. What performance measure is used during training?

The primary performance measure is the **episode reward** and its **average reward over a specified averaging window**. In this experiment, training stopped when the average reward reached approximately -3.

24. How does the agent's performance change as training progresses?

Initially, the performance is highly variable and rewards are generally low. As training progresses, the average reward improves and gradually approaches the stopping threshold. Eventually, the performance stabilizes around an average reward of -3, indicating that the agent has learned an effective policy.

Task 3: Learning Curve Observation Table

The ranges below are based on your actual learning curve, which ran for **886 episodes**.

Training Stage	Episode Range	Reward/Performance	Observation
Initial	1–50	Highly variable; some rewards reach approximately -20	The agent is mostly exploring and has not yet learned an effective strategy
Early Training	51–200	Rewards remain variable, but the average reward begins improving	The agent starts learning from interactions and improves its action selection
Middle Training	201–500	Average reward gradually moves closer to -3 to -4	The learned policy improves, although individual episode rewards still fluctuate
Late Training	501–750	Average reward remains relatively stable near the stopping threshold	The agent shows more consistent performance and learning begins to stabilize
Final	751–886	Final Average Reward = **-2.99**	The agent reaches the stopping criterion and training automatically terminates at episode 886

Task 3: Analysis Questions

25. What does the learning curve represent?

The learning curve represents how the RL agent's performance, usually measured using episode reward and average reward, changes over the course of training episodes.

26. Why is the reward generally low during the initial training episodes?

During the initial episodes, the agent has little or no knowledge about which actions are optimal. It explores different actions randomly or according to its exploration policy, which can result in poor decisions and low rewards.

27. What indicates that the agent is learning?

An increase or improvement in the average reward over time indicates that the agent is learning. In the learning curve, the average reward gradually moves toward the stopping threshold and eventually reaches approximately -3.

28. Does the reward increase consistently throughout training? Explain.

No. The reward does not increase consistently in every episode. Reinforcement learning involves exploration, and different episodes can produce different outcomes. Therefore, individual rewards fluctuate even when the overall average performance is improving.

29. How can you identify convergence from the learning curve?

Convergence can be identified when the average reward stabilizes and no longer shows significant long-term improvement. In this experiment, convergence was indicated when the average reward reached the stopping value of approximately **-3**, causing training to stop at episode **886**.

30. What could cause fluctuations in the learning curve?

Fluctuations can occur because of:

- Exploration of different actions
- Randomness in the environment
- Different states encountered during episodes
- The agent trying actions that are not optimal
- Variations in episode outcomes

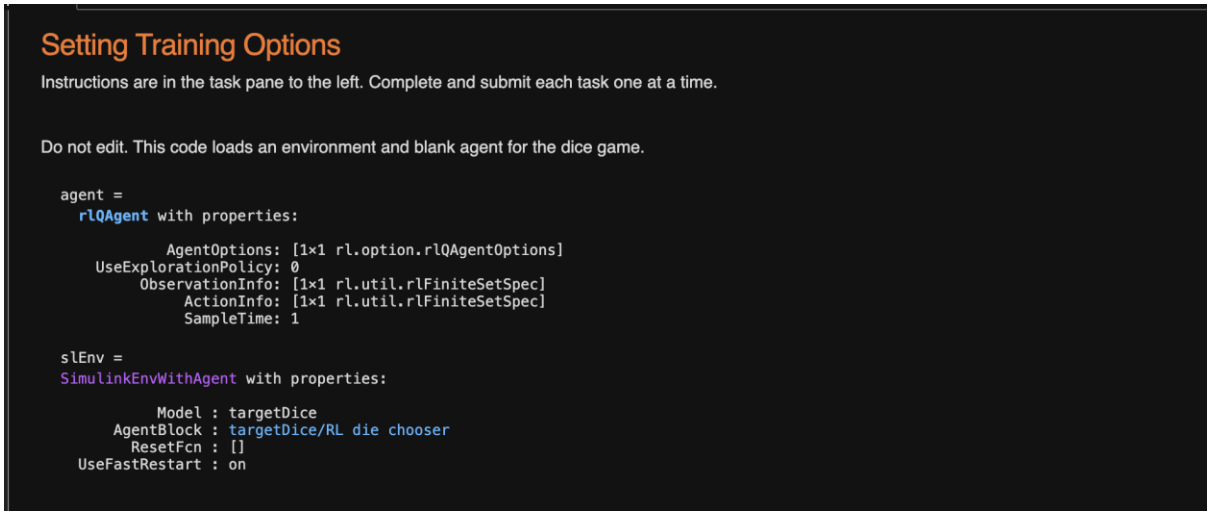
These fluctuations are normal during reinforcement learning.

31. What happens if the agent is trained for a larger number of episodes?

Training for more episodes can potentially improve the policy further if the agent has not fully converged. However, after convergence, additional training may provide little improvement and mainly increase computational time. Performance may also continue fluctuating around a stable value.

32. Why is it important to analyze the learning curve rather than considering only the final reward?

The final reward represents only one episode and may not accurately reflect the overall performance of the agent. The learning curve shows the complete learning process, including initial performance, improvement, fluctuations, stability, and convergence. Therefore, it provides a more reliable evaluation of the agent's training behaviour.







Task 2

```
diceopts =
  rlTrainingOptions with properties:

                  MaxEpisodes: 2000
            MaxStepsPerEpisode: 20
                   StopOnError: "on"
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        SaveSimulationDirectory: "savedSims"
                 SaveFileVersion: "-v7"
      ScoreAveragingWindowLength: 5
           StopTrainingCriteria: "AverageSteps"
              StopTrainingValue: 500
              SaveAgentCriteria: "none"
                SaveAgentValue: "none"
             SaveAgentDirectory: "savedAgents"
                        Verbose: 0
                          Plots: "training-progress"
                    UseParallel: 0
         ParallelizationOptions: [1×1 rl.option.ParallelTraining]
```

Task 3

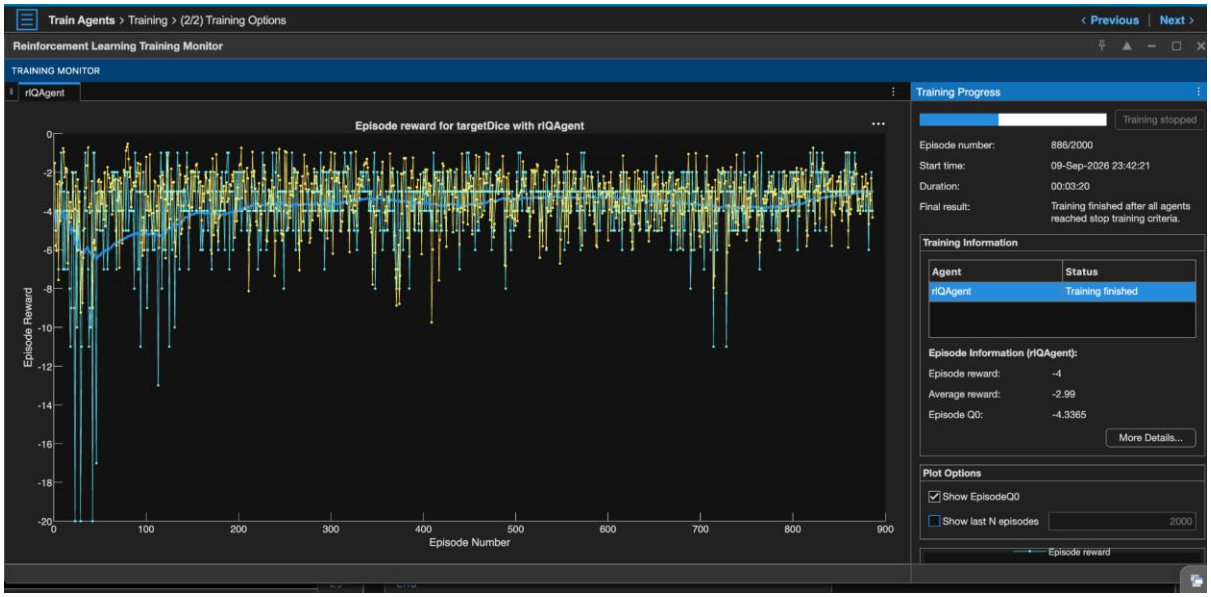
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                 SaveFileVersion: "-v7"
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           StopTrainingCriteria: "AverageSteps"
              StopTrainingValue: 500
              SaveAgentCriteria: "none"
                SaveAgentValue: "none"
             SaveAgentDirectory: "savedAgents"
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                          Plots: "training-progress"
                    UseParallel: 0
         ParallelizationOptions: [1×1 rl.option.ParallelTraining]
```

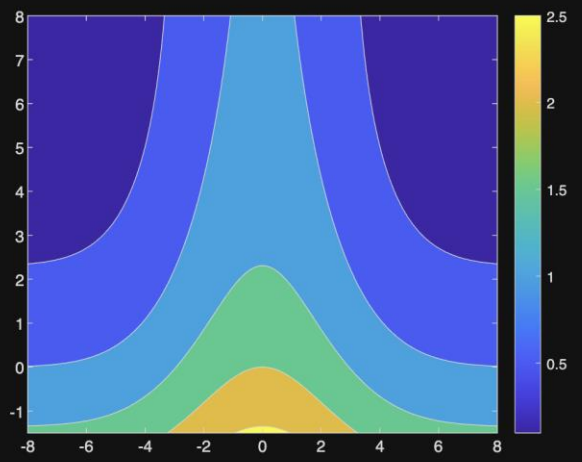
Task 4

```
diceopts =
  rlTrainingOptions with properties:

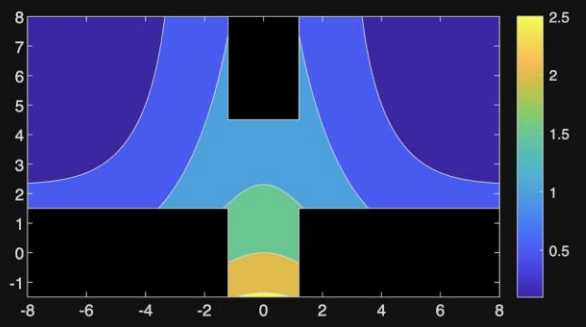
                    MaxEpisodes: 2000
              MaxStepsPerEpisode: 20
                     StopOnError: "on"
           SimulationStorageType: "memory"
        SaveSimulationDirectory: "savedSims"
                 SaveFileVersion: "-v7"
     ScoreAveragingWindowLength: 100
           StopTrainingCriteria: "AverageReward"
              StopTrainingValue: -3
              SaveAgentCriteria: "none"
                 SaveAgentValue: "none"
              SaveAgentDirectory: "savedAgents"
                        Verbose: 0
                          Plots: "training-progress"
                    UseParallel: 0
          ParallelizationOptions: [1×1 rl.option.ParallelTraining]
```



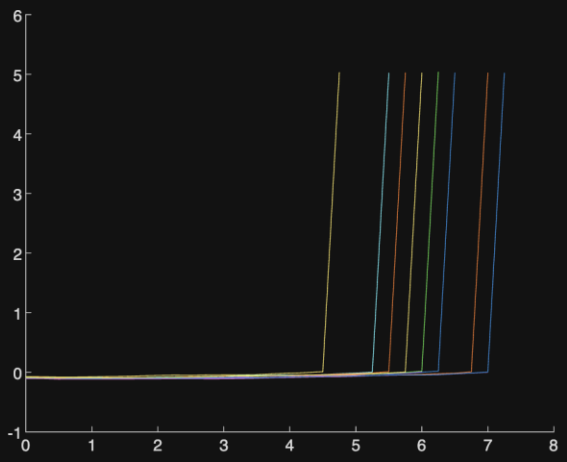
Task 3



Do not edit. This code adds a visualization of the environment obstacles.



Task 4



Task 3

